

Summary of Scenario and Model

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I. INTRODUCTION OF SCENARIO

A. Introduction of physical elements in the delivery system

The components of the UAV delivery system can be described as follows: a warehouse, which can be denoted as WH, is the place with lots of parcels that are waited to be delivered. The delivery company will have a warehouse in each suburb and the parcels concentrated in this warehouse will be delivered to the customers who lives in this suburb. A delivery UAV, which can be denoted as Ux, will carry a package from the warehouse and send the package to the customer's home. On the way of the delivery, two different types of signal base stations will be structured as arrays and can be used to guide and supervise the delivery process of UAVs. The aerial base station can be marked as Ax Node. Those aerial base stations are Helium balloon airship characteristics with ropes traced and fixed on the ground and can be used to communicate with delivery UAVs directly. The ground base station can be marked as Gx Node. Those ground base stations are a part of ground-based infrastructure and are mainly used to receive and transfer data and message between delivery UAVs and warehouse. Both aerial base stations and ground base stations will use Wi-Fi to communicate with each other and the delivery UAVs will apply Bluetooth as the communication tool with aerial base stations. The effective connection distance for Wi-Fi will be over 100 meters, but it can also cause battery failure for UAV since UAV needs power for long-distance flying. Bluetooth is much more power-friendly for UAV but the effective connection range is only about 10 meters. Thus, the Bluetooth receivers are also applied on aerial base stations so deliver UAVs can communicate with the base station system. The customer will use home Wi-Fi with Bluetooth receiver, which can be marked as Hx Node, to make the delivery UAV get connected with the MANET to receive parcels and further operations. We can assume that the deployment of aerial base stations and delivery UAVs are much higher than buildings in smart city so that collision problems and avoiding obstacles can be ignored for accurate movement.

B. Introduction of the delivery process

This figure shows the structure of UAV delivery system. The delivery UAV will start from the warehouse carried with the parcel and a list of aerial base station id. The list of aid is provided by warehouse and shows the aerial base stations that the delivery UAV will pass to reach the destination of customer's home. As an example, the UAV U1 need to deliver the parcel to the home H1, it needs to reach and pass A4, A5, then Ax until it successfully reaches H1. Thus, the computer in warehouse will read the list of aid and lead U1 to the

first location A4. Basically, both warehouse and base stations will be connected to GPS and have their own coordinate points (x,y) displayed in the system. It is clear that if a node has the coordinate point for another node, it can easily calculate the direction and the distance between these two nodes. The delivery UAV can receive direction and distance information and use them to fly from one node to the specified node accurately. Back to the situation, WH will calculate the direction and distance of A4 and send them to U1, so U1 can follow the guide and reach A4. Once U1 reaches and communicates with A4, U1 will send the aid of A5 from the list to A4 and A4 will start the process of ProcessFindNode to connect and find the location of A5. This process will be introduced in ω -calculus in the next subsection. When A5 received the request of finding node, it will send its coordinate location back to A4. A4 will then calculate the direction and distance towards A5 and send these to U1. Then, U1 will follow the direction and distance to start moving from A4 and reach A5. After U1 reaches and get connected with A5, it will send A5 the next aid from the list and A5 will start the ProcessFindNode to search the location of the next node. This process will be circulated until it reaches H1. The customer will receive the parcel with a series of receiving process and after delivery is finished, the UAV will reverse the aerial base station node in the AList and follow by the process again until it reaches WH.

C. Specification of the node finding process in the ω -calculus

There are three different kinds of devices in this system: the delivery UAV device Ux, the aerial base station Ax and the ground base station Gx, that can be regarded as a "node". Each node runs an instance of process node(id, isFound, init, find, tid, pChan, WHChan). In this definition, id is the node identifier and each type of nodes will have different types of id. We are going to mark "uid" for UAV devices, "aid" for aerial base station devices and "gid" for ground base station devices. isFound is a Boolean parameter which indicates whether the node with the target id has been found in the finding process or not. init is a Boolean parameter which indicates whether the node initiates the finding process. find is a Boolean indicates whether the node has participated in this finding process. tid is the id of the target node. pChan indicates the parent node that send the finding message to it. WHChan is a special dedicated channel for nodes to get communication with the warehouse WH.

A node may receive five kinds of messages during ProcessFindNode:

- Finding message: the message will be sent from the delivery UAV to the aerial base station in its communication

range. Once an aerial base station received this message from a UAV, it will become the start node and initiate the finding process. The finding message will carry the parameter tid and the init aerial base station will broadcast the finding message with tid to other base stations to find the target node.

- Acknowledge message: this message is an acknowledgment to the finding process
- Location message: this message is the position of the target node that UAV needs. Once the target aerial base station found, it will send this location message back to the start node. This location message will contain the coordinate point displayed in the GPS.
- NodeNotFound message: after every node has been reached but no target node found, the start node will send this error message with uid back to WH to report this error.
- UAVState message: after the UAV received the direction and the destination and start the movement to the next aerial base station, the start node will send this message with uid back to WH so that WH can keep track on the movement and state of delivery UAVS.

Thus, we can determine the instance of node process using ω -calculus:

$$\begin{aligned} node(id, isFound, init, find, tid, pChan, WHChan) \stackrel{\text{def}}{=} \\ r(finding(tid)).processFindNode(id, \\ isFound, init, find, tid, pChan, WHChan) \\ + r(ack(tid), location).processAck(id, \\ isFound, init, find, tid, pChan, WHChan) \end{aligned}$$

In this calculus, $r(finding(tid))$ means the node receives the finding message with the input tid. Then, it reaches the processFindNode state where it broadcast the finding message and goes to state awaitAck.

$$\begin{aligned} processFindNode(id, isFound, init, find, tid, pChan, \\ WHChan) \stackrel{\text{def}}{=} \\ [id! = tid]b(finding(tid)).awaitack(id, \\ isFound, init, find, tid, pChan, WHChan) \\ + [id == tid]pChan(ack(tid), location).processAck(id, \\ isFound, init, find, tid, pChan, WHChan) \end{aligned}$$

In the processFindNode, the start node will broadcast the finding message with tid to other nodes that can communicate with it directly. Each node will receive that finding message and mark the start node as their parent node and keep the channel of the start node as pChan. They will compare their own id with the tid in the finding message. If it matches, it will unicast the acknowledge message and location message to its parent node stored in pChan and then reach to processAck with the Boolean isFound = true. If it does not match, it will broadcast the finding message with the tid and then reach to the state awaitAck.

A node in awaitAck state may receive an ack message and reach processAck state or cannot reach processAck state if every node is traversed but no one matches. In the previous case, if the init Boolean is false, that is, the node is not the start node, the node will unicast the ack message with tid and location message to its parent node via pChan until the messages reach the start node. If the init == true, that is, the node is the start node, the start node will calculate the distance and direction between start node and target node using GPS coordinate point and unicast the calculated location message to the delivery UAV. Then it will reach the processReportState to report the state of delivery UAV to WH. In the latter case, the start node will unicast the error message back to WH using WHChan, which will be discussed later in this section.

$$\begin{aligned} awaitAck(id, isFound, init, find, tid, pChan, WHChan) \stackrel{\text{def}}{=} \\ [init == false]pChan(ack(tid), location).processAck(id, \\ isFound, init, find, tid, pChan, WHChan) \\ + [init == true]pChan(calculatedlocationmessage) \\ .processReportState(id, uid, WHChan) \end{aligned}$$

In the processReportState, the start node will send UAV state message back to WH.

$$\begin{aligned} processReportState(id, uid, WHchan) \stackrel{\text{def}}{=} \\ WHChan(UAVState(uid)).endProcessFindNode() \end{aligned}$$

A node can broadcast the finding message with tid to all nodes connect with it with the find == false. If no node can receive the finding message, it is clear that this node is at the end of this spanning tree and will report a notification nodeNotFound back to the start node. If the start node received every nodeNotFound message from the leaf of the spanning tree but still no location message received, it will reach the processReportErrors and report the error message nodeNotFound to WH.

$$\begin{aligned} processReportError(id, uid, WHChan) \stackrel{\text{def}}{=} \\ WHChan(NodeNotFound(uid)).endProcessFindNode() \end{aligned}$$